DOI: 10.1111/ajag.12159

Research

Depression screening among older adults attending low-vision rehabilitation and eye-care services: Characteristics of those who screen positive and client acceptability of screening

Edith E. Holloway and Bonnie A. Sturrock
Centre for Eye Research Australia, Royal Victorian Eye and Ear Hospital, University of Melbourne, Melbourne, Victoria, Australia

Ecosse L. Lamoureux
Centre for Eye Research Australia, Royal Victorian Eye and Ear Hospital, University of Melbourne, Melbourne, Victoria, Australia; Singapore Eye Research Institute, National University of Singapore; and Duke–National University of Singapore Graduate Medical School, Singapore

Jill E. Keeffe and Gwyneth Rees
Centre for Eye Research Australia, Royal Victorian Eye and Ear Hospital, University of Melbourne, Melbourne, Victoria, Australia

Aim: *To investigate characteristics associated with screening positive for depressive symptoms among older adults accessing low-vision rehabilitation and eye-care services and to determine client acceptability of depression screening using the Patient Health Questionnaire-2 (PHQ-2) in these settings.*
Methods: *One-hundred and twenty-four older adults (mean = 77.02 years, SD = 9.12) attending low-vision rehabilitation and eye-care services across Australia were screened for depression and invited to complete a telephone-administered questionnaire to determine characteristics associated with depressive symptoms and client acceptability of screening in these settings.*
Results: *Thirty-seven per cent (*n = 46/124*) of participants screened positive for depressive symptoms, and the majority considered the new depression screening method to be a 'good idea' in vision services (85%). Severe vision loss (<6/60 in the better eye) was associated with an increased odds of screening positive for depressive symptoms (odds ratio 2.37; 95% confidence interval 1.08–6.70) even after adjusting for potential confounders. Participants who screened positive had a preference for 'talking' therapy or a combination of medication and 'talking therapy' delivered within their own home (73%) or via telephone (67%).*
Conclusion: *The PHQ-2 appears to be an acceptable method for depression screening in eye-care settings among older adults. Targeted interventions that incorporate home-based or telephone delivered therapy sessions may improve outcomes for depression in this group.*

Correspondence to: Dr Gwyneth Rees, Centre for Eye Research Australia, Royal Victorian Eye and Ear Hospital, University of Melbourne. Email: grees@unimelb.edu.au

Key words: *depression screening, low vision, psychological treatment preference, vision rehabilitation.*

Introduction

The link between vision impairment and depression in older adults is well established, and studies have shown that higher levels of vision impairment may be associated with increased risk [1]. For a person with vision impairment, depressive symptoms can lead to increased levels of disability and functional decline and a greater risk of developing major depression, which may act as a barrier to effective vision rehabilitation outcomes [2,3]. Early identification and management of depressive symptoms in this group is therefore critical.

The need to identify depression in high-risk groups has been emphasised in screening guidelines, and the UK's National Institute of Clinical Excellence (NICE) has recommended routine depression screening for people with functional impairment within health-care settings [4]. Studies conducted in health conditions including epilepsy have yielded support for depression screening [5] and have generally found a high rate of patient acceptability, with a preference for brief screening tools [6,7].

In low-vision rehabilitation services throughout Australia, staff have undertaken training [8] to screen for depressive symptoms using the Patient Health Questionnaire-2 (PHQ-2) [9], a two-item measure that has been validated in previous research and is widely used as a first-line screening instrument. The PHQ-2 has recently demonstrated its clinical utility in oncology settings [10] and with coronary heart disease patients [11]. It has also demonstrated good sensitivity, specificity and ease of administration for older adults when compared with 'gold-standard' diagnostic interviews [12].

While depression screening is the first step to ensuring patients gain access to treatment through appropriate referrals, screening alone is not sufficient to improve outcomes [13], and greater emphasis has been placed in current screening guidelines on the importance of combining screening with effective treatment and follow-up [14]. Adherence to treatment in older adults is frequently reported to be poor [15]; however, evidence suggests that patients who receive their preferred treatment show greater improvements and adherence to treatment compared to those who do not [16]. A better understanding of treatment preferences in this group is therefore needed.

© 2014 ACOTA

The primary aims of this exploratory study were the following: (i) to determine the proportion and characteristics of participants who screened positive for depressive symptoms in a sample of older adults accessing low-vision rehabilitation and eye-care services; and (ii) to determine client acceptability of depression screening using the PHQ-2 in these settings. A secondary aim was to identify preferences for psychological support services for those who screened positive.

Methods

Study design and setting

Participants were recruited from Vision Australia centres across Victoria, New South Wales, Queensland and the Australian Capital Territory and the Royal Victorian Eye and Ear Hospital (RVEEH) between November 2010 and July 2012 as part of a prospective longitudinal study exploring outcomes of depression screening [17]. This study reports on findings from the baseline assessment.

Participants and procedure

To be eligible for the study, participants were required to have visual acuity <6/12 in the better eye (best corrected), be aged 60 years or older, be able to converse in English, have adequate hearing (with or without a hearing aid) and have no cognitive impairment as determined by a score of ≤7 on the six-item Cognitive Impairment Test (6CIT) [18].

Potential participants who met visual acuity and age eligibility criteria were identified by Vision Australia staff during the new intake assessment, in which the PHQ-2 [9] was administered by Vision Australia staff as a standard component of intake. Following depression screening, consent to be contacted by the research team was obtained by Vision Australia staff. Consenting clients were contacted by the research team, who completed a cognitive assessment. At the RVEEH site, patient records were reviewed by a research assistant who completed the PHQ-2 with eligible patients waiting for their clinic appointment. Across all sites, clients who screened positive for depressive symptoms were advised that their responses warranted further follow-up and were offered a referral letter to discuss depression with their GP. Educational materials (e.g. *beyondblue* resources) about depression and services available in the community were also offered. All eligible clients were invited to take part in the study, and a baseline assessment was scheduled within 1 week of depression screening with consenting clients. A research assistant completed the baseline assessment via telephone to determine demographic characteristics, acceptability of depression screening and preferences for psychological support services.

Ethical approval was received from the Human Research and Ethics Committee of the RVEEH (project no. 09/923H). Written consent was obtained from all eligible participants, including permission to access ophthalmology and low-vision rehabilitation reports.

Measures

Background

Age, gender, marital status, country of birth, languages spoken, education level and general health were recorded via self-report at the baseline assessment. Participants were also asked to self-report if they had ever suffered from depression ('no', 'yes, in the past', 'yes, currently') and if they had ever received treatment for depression in the form of antidepressant medication, therapy or counselling ('no', 'yes, in the past', 'yes, currently').

Ocular health

The main cause of vision loss and visual acuity were extracted from Vision Australia and the RVEEH patient files. Visual acuity was assessed using logarithm of the minimum angle of resolution (LogMAR) charts at Vision Australia and the Snellen chart at RVEEH. Visual impairment was categorised as mild (<6/12–6/18), moderate (<6/18–6/60) or severe (<6/60).

Depressive symptoms

Participants were screened for depression using the PHQ-2 [9]. The PHQ-2 consists of the first two items of the PHQ-9 [19] (see Appendix I), which assess depressed mood and loss of interest (anhedonia) over the previous 2 weeks. At least one of these symptoms is required to establish a diagnosis of major depressive disorder (MDD) or any depressive disorder listed in the *DSM-IV*. PHQ-2 scores can range from 0 to 6, and a score of ≥3 suggests clinically significant depression, which should prompt either completion of the full PHQ-9 or a clinical interview to assess for MDD [20]. A score of ≥3 has been shown to have sensitivity as high as 83% and specificity of 92% for major depression [9]. The PHQ-2 has been validated in previous research for use in older adults [12].

Acceptability of depression screening

The Patient Acceptability Scale (PAS) [21] was used to determine the extent to which participants felt 'annoyed', 'embarrassed', 'upset' or 'uncomfortable' in answering the two-question screening tool. Response options ranged from 1 (very much) to 4 (not at all). Participants were also asked to indicate if they thought it was a good idea to screen for depression ('yes', 'no', 'unsure').

Perceived need for psychological help

Participants were asked to indicate the extent to which they felt the need for help on a single item rated on a five-point Likert scale. A higher score indicated a greater perceived need for psychological help.

Preferences for psychological treatments

To determine preferences for psychological treatment, participants who screened positive for depressive symptoms were asked to indicate (i) which of four treatment options was their first preference, (ii) which was their second preference and (iii) if any were unacceptable choices [22]. All participants, regardless of their treatment preference, were

Australasian Journal on Ageing, Vol 34 No 4 December 2015, 229–234
© 2014 ACOTA

asked to consider five modes of delivery for receiving 'talking' therapy from a psychologist or counsellor and whether they felt it was an acceptable or unacceptable setting in which to receive therapy.

Analysis

Statistical analysis was carried out using SPSS version 19.0. The cut-off score for the PHQ-2 was ≥3, indicating depressive symptoms warranting further clinical attention (i.e. a positive screen). The Kolmogorov–Smirnov test of normality was statistically significant ($P < 0.05$), indicating that the data were not normally distributed and non-parametric tests were warranted. Chi-square and non-parametric tests (e.g. Mann–Whitney *U*) were used to compare between-group differences (positive versus negative depression screen) for each of the sociodemographic variables. To identify factors associated with depressive symptoms, exploratory analyses were performed. Variables identified as associated with the outcome in chi-square and non-parametric tests ($P < 0.05$) were entered in univariate regression models. A multivariate logistic regression model was performed to determine if level of visual acuity remained associated with depressive symptoms after adjustment for potential confounders. Odds ratios and 95% confidence intervals for a positive depression screen relative to a negative screen are reported.

Results

Participants and response rates

The files of 234 clients were reviewed for visual acuity, of whom 33 (14%) did not meet the eligibility criteria. Of the eligible clients, 42 (21%) declined to be screened for depression, three (2%) did not meet other eligibility criteria and a further 32 (16%) were screened but declined to take part in the study; of the latter, 41% ($n = 13$) screened positive for depressive symptoms. In this paper, we report on the baseline data from 124 eligible participants.

The participants ranged in age from 60 to 94 years (mean = 77.02, SD = 9.12); most were female ($n = 88$, 71%) and reported having at least one chronic health condition besides their eye condition (86%; Table 1). The most prevalent causes of vision loss were age-related macular degeneration (51%) and glaucoma (18%), and the majority (83%) had mild or moderate vision loss.

Proportion and characteristics of participants who screened positive for depressive symptoms

Thirty-seven per cent ($n = 46/124$) of participants screened positive for depressive symptoms, and no differences were found in the proportion who screened positive at Vision Australia centres ($n = 38/106$; 36%) and the proportion who screened positive at the RVEEH ($n = 8/18$; 44%) ($P = 0.33$). Participants who screened positive were more likely to have severe vision loss ($\chi^2(df = 2, n = 124) = 4.73$, $P = 0.04$), have current depression ($\chi^2(df = 2, n = 124) = 49.71$, $P > 0.001$) and currently be receiving treatment for depression ($\chi^2(df = 2, n = 124) = 37.55$, $P > 0.001$) compared to those who screened negative (Table 1). Overall, 22% ($n = 10/46$) of participants who screened positive did not report a history of depression, and 57% ($n = 26/46$) were not currently receiving treatment. Participants who screened positive reported a greater need for psychological support than those who screened negative ($z = -3.64$, $P < 0.001$).

Univariate logistic regression analyses showed that participants with severe vision loss had an estimated 2.37-fold increase in the odds of screening positive for depressive symptoms compared to those with mild vision loss (95% CI 1.08–6.70; $P = 0.04$) (Table 2). This association remained significant in a multivariate model after controlling for age, gender, history of depression, treatment status and perceived need for psychological support (OR 7.41; 95% CI 1.59–34.62; $P < 0.001$). History of depression, treatment status and a perceived need for psychological support were also associated with a positive depression screen in univariate analyses (all $P < 0.05$; Table 2).

Acceptability of depression screening

The majority of participants believed it was a 'good idea' to screen for depression at eye-care and vision rehabilitation services (85%). Most participants rated the questions of the PHQ-2 as 'easy' (45%) or 'very easy' (41%) to answer and reported not feeling embarrassed (96%), upset (96%), uncomfortable (96%) or annoyed answering the questions. Participants who screened positive did not differ in their level of acceptance towards depression screening compared to those who screened negative (all $P > 0.05$).

Preferences for emotional support and psychological treatments

Participants who screened positive for depressive symptoms ($n = 46$) preferred 'talking therapy' from a counsellor or psychologist (29%) or a combination of medication and 'talking therapy' (33%). Fewer participants had a preference for antidepressant medication (20%) or no treatment (18%). In general, participants were accepting of most settings for receiving 'talking therapy', including at home (73%) and via telephone (67%). Around half reported they would be happy to receive sessions in a therapist's office (56%) or a Vision Australia centre (46%), while only 7% were accepting of sessions over the Internet.

Discussion

This is the first study to report on the acceptability of depression screening at low-vision rehabilitation and eye-care services among older adults. On a brief, well-validated screening tool, administered largely by trained low-vision rehabilitation staff, over a third of participants scored above the clinical cut-off for depressive symptoms, warranting further clinical attention. Encouragingly, many participants indicated little negative affect following administration of the screening questions. Over half of those participants who screened positive for depressive symptoms were not currently

© 2014 ACOTA

Table 1: Sociodemographic characteristics of participants by PHQ-2 score

Characteristic	Total ($n = 124$)	Positive screen ($n = 46$, 37.1%)	Negative screen ($n = 78$, 62.9%)	*P* value‡
Age (years), mean ± SD (range)	77.02 ± 9.12 (60–94)	75.18 ± 8.80	78.12 ± 9.19	0.09
Sex, *n* (%)				
Female	88 (71.0)	34 (38.6)	54 (61.4)	0.37
Male	36 (29.0)	12 (33.3)	24 (66.7)	
Country of birth				
Australia	99 (79.8)	35 (35.4)	64 (64.6)	0.81
Other	25 (20.2)	11 (44.0)	14 (56.0)	
Marital status, *n* (%)				
Married/de facto/partner	56 (45.2)	20 (35.7)	36 (64.3)	0.67
Widowed	48 (38.7)	20 (41.7)	28 (58.3)	
Divorced/separated	14 (11.3)	5 (35.7)	9 (64.3)	
Never married	6 (4.8)	1 (16.7)	5 (83.3)	
Level of education, *n* (%)				
Some secondary/technical education	54 (43.5)	15 (27.8)	39 (72.2)	0.12
None/primary school equivalent	31 (25.0)	15 (48.4)	16 (51.6)	
Secondary school completed	16 (12.9)	5 (31.3)	11 (68.7)	
Trade/apprenticeship/some college/technical and further education	14 (11.3)	5 (35.7)	9 (64.3)	
Degree qualified	9 (7.3)	6 (66.7)	3 (33.3)	
Comorbid physical illness, *n* (%)				
Yes	106 (86.2)	41 (38.7)	65 (61.3)	0.18
No	17 ((13.8)	4 (23.5)	13 (76.5)	
Visual acuity, *n* (%)				
Mild (<6/12–6/18)	50 (40.3)	18 (36.0)	32 (64.0)	0.04
Moderate (<6/18–6/60)	53 (42.7)	16 (30.2)	37 (69.8)	
Severe (<6/60)	21 (16.9)	12 (57.1)	9 (42.9)	
Eye condition†				
Macular degeneration	63 (51.2)	21 (33.3)	42 (66.7)	0.60
Other	24 (19.5)	11 (45.8)	13 (54.2)	
Glaucoma	22 (17.9)	8 (36.4)	14 (63.6)	
Diabetic retinopathy	11 (8.9)	3 (27.3)	8 (72.7)	
Cataracts	3 (2.4)	2 (66.7)	1 (33.3)	
History of depression, *n* (%)				
No, never	74 (59.7)	10 (13.5)	64 (86.5)	<0.001
Yes, currently	29 (23.4)	25 (86.2)	4 (13.8)	
Yes, in the past	21 (16.9)	11 (52.4)	10 (47.6)	
Previous treatment, *n* (%)				
No, never	75 (60.5)	12 (16.0)	63 (84.0)	<0.001
Yes, currently	26 (21.0)	20 (76.9)	6 (23.1)	
Yes, in the past	23 (18.5)	14 (60.9)	9 (39.1)	
Perceived need for help, mean ± SD	2.61 ± 1.05	3.09 ± 1.19	2.33 ± 0.85	<0.001

† One participant had missing data. ‡ χ^2-test, Mann–Whitney *U*-test. PHQ-2, Patient Health Questionnaire-2.

Table 2: Univariate and multivariate associations between predictor variables and a positive depression screen (score ≥3) on the PHQ-2

Variable	Unadjusted OR	95% CI	Adjusted OR	95% CI
Age	0.96	0.93–1.00	0.95	0.89–1.00
Gender	1.25	0.56–2.84	0.49	0.12–1.92
Visual impairment				
Mild	1.00 (ref)		1.00 (ref)	
Moderate	0.77	0.34–1.75	0.60	0.18–1.96
Severe	2.37	1.08–6.70*	7.41	1.59–34.62*
History of depression				
No, never	1.00 (ref)		1.00 (ref)	
Yes, in the past	7.04	2.38–20.84**	4.76	0.61–36.95
Yes, currently	40.00	11.48–139.37**	69.73	6.43–756.64**
Previous treatment				
No, never	1.00 (ref)		1.00 (ref)	
Yes, in the past	8.17	2.89–23.11**	3.63	0.49–26.70
Yes, currently	16.50	5.82–52.65**	16.50	0.09–8.28
Perceived need for psychological help	2.02	1.40–2.94**	1.37	0.81–2.32

*$P < 0.05$; **$P < 0.01$; ***$P < 0.001$. PHQ-2, Patient Health Questionnaire-2; ref, reference group; OR, odds ratio; CI, confidence interval.

© 2014 ACOTA

receiving treatment for depression, underlining the importance of screening and referrals in these settings.

The frequency of patients who screened positive for depressive symptoms in our sample was slightly higher than that reported in other low-vision samples [2,23]; however, those data were often gathered in general community rather than eye-care settings. In low-vision rehabilitation settings, seminal work examining the frequency of depression using a formal diagnosis found that 34% of older adults (versus 37% who screened positive for depressive symptoms in our study) met the criteria for major or subthreshold depression [3]. In the current study, 86% of participants who screened positive for depressive symptoms also self-reported current depression, while 87% who scored <3 reported no current diagnosis. Taken together, these results provide information about the plausibility of a positive screen and point to the PHQ-2 as a useful first-stage screening tool. However, as we did not include a formal diagnosis of depression, we cannot know with any certainty how many false positive and false negative screens are among our data.

We found that poorer visual acuity was associated with a positive depression screen. It may be the level of disability associated with the vision loss that increases the risk of depression rather than visual impairment itself. For example, a recent study using a probability-based sample showed that physical limitations and social isolation were the mechanisms by which older adults developed depression, irrespective of the degree of vision loss [24].

The majority of participants who screened positive preferred 'talking' therapies, rated as the treatment of choice alone or in combination with antidepressants. This is consistent with research demonstrating that psychological treatments for depression are favoured over antidepressant medication in older adults [22]. Psychological therapies have also been shown to be a more effective treatment strategy for depressive symptoms that do not meet the full diagnostic criteria, known as subthreshold depression, levels of which are commonly experienced by vision-impaired adults [3]. Our findings showed that two-thirds of participants who screened positive would be happy to receive 'talking' therapy over the phone. Due to barriers such as lack of available and accessible services, transportation problems and cost, treatment adherence to psychological therapies is often reported to be low in primary care [25]. Telephone-administered therapy in addition to face-to-face sessions via home visits or video conferencing, particularly important for those with hearing difficulties, may provide suitable delivery modes to older vision-impaired adults for whom travel is a challenge.

Several limitations and caveats exist in interpreting these findings. Previous studies have shown depression to be a barrier to seeking low-vision services [3], and as such, the rates of depression and severity of symptoms may be lower in this sample, which was largely recruited from low-vision services. As noted, the percentage of positive screens was also quite high in those who declined to participate in the research. This may leave a knowledge gap in terms of the characteristics of those clients not completing the PHQ-2 (hence not invited to take part in the study) and how these differed from those of the invited clients. Furthermore, as the majority of our sample had not yet received low-vision rehabilitation services, we did not assess service use. Future research in this area should determine the impact of rehabilitation services on depressive symptoms. Due to the small sample size, we were unable to stratify the data exploring treatment preference by exposures such as history of depression, current treatment and perceived need for psychological help, as these factors may also influence treatment preferences. Individuals with a cognitive impairment were excluded from this study; however, the co-occurrence of depression and cognitive impairment among older adults is common, highlighting the importance of identifying depression in this group. Vision rehabilitation and eye-care providers should also consider training staff in the use of depression screening tools that have been validated for use in individuals with dementia. These instruments generally require more time to administer than briefer tools such as the PHQ-2 [26].

Despite these limitations, this study provided a novel method for identifying depression that involved internal screening by trained staff of eye-care and low-vision rehabilitation providers. This study was not designed to evaluate the psychometric properties of the PHQ-2, and the current presence of depression was determined via self-report. Future studies could provide data on the clinical utility of the PHQ-2 compared to a 'gold-standard' instrument for use in older adults with vision impairment. The positive predictive value of a PHQ-2 score ≥3 has been reported to be around 34% [27]. Therefore, eye-care providers could also consider a two-step approach adopted by the American Heart Association (AHA) Prevention Committee, where the PHQ-9 [19] is administered for confirmation of a positive PHQ-2 result [28]. An important benefit of this two-stage approach is that many false-positive PHQ-2 screen results could be eliminated, as has been shown in previous studies [29].

In conclusion, this study provides evidence to support acceptability among older adults of depression screening using a brief tool, the PHQ-2, administered by staff at eye-care and low-vision rehabilitation providers. The findings from this study provide preliminary data on the types (and modes of delivery) of treatment that would be acceptable to vision-impaired older adults. These preferences may be useful in informing the development of appropriate and effective treatment to be used in conjunction with screening tools. We are currently integrating an evidence-based treatment for depression (Problem-Solving Therapy for Primary Care, PST-PC) into low-vision rehabilitation settings [30]. Staff will be trained to deliver PST-PC via telephone to those clients who screen positive for depressive symptoms on the PHQ-2 and, subsequently, the PHQ-9. Integrating such services into low-

© 2014 ACOTA

vision rehabilitation care could help to overcome some of the practical and logistical barriers to the uptake of referrals for emotional support to outside agencies, allow emotional and rehabilitation needs to be addressed simultaneously, and enable greater continuity of care.

Acknowledgements

We would like to thank beyondblue – *the National Depression Initiative – which provided funding to support this study, and all participants who took part. This project was funded by* beyondblue – *the National Depression Initiative. We gratefully acknowledge the contribution from Michael Kienhuis, Sam Coad, Niky Hepi and Graeme Craig at Vision Australia in assisting with coordinating this study within Vision Australia. Gwyneth Rees is currently a NHMRC Translating Research Into Practice Fellow (2012–2014). The Centre for Eye Research Australia receives Operational Infrastructure Support from the Victorian Government and the Centre for Clinical Research Excellence.*

Key Points

- Older adults with vision impairment are at increased risk of depression, and methods to improve the detection of depression are needed.
- The study provides evidence to support the acceptability of depression screening in low vision rehabilitation and eye-care services among older adults.
- Depression screening should be combined with appropriate treatment, and future studies should investigate such services in low vision rehabilitation settings.

References

1 Brown RL, Barrett AE. Visual impairment and quality of life among older adults: An examination of explanations for the relationship. *The Journals of Gerontology. Series B, Psychological Sciences and Social Sciences* 2011; 66B: 364–373.
2 Rovner B, Ganguli M. Depression and disability associated with impaired vision: the MoVies Project. *Journal of the American Geriatrics Society* 1998; 46: 617–619.
3 Horowitz A, Reinhardt JP, Kennedy GJ. Major and subthreshold depression among older adults seeking vision rehabilitation services. *The American Journal of Geriatric Psychiatry* 2005; 13: 180–187.
4 National Collaborating Centre for Mental Health. Depression in adults with a chronic physical health problem: Treatment and management. London: National Institute for Clinical Excellence, 2009.
5 Margrove K, Thapar AK, Mensah SA, Kerr MP. Help-seeking and treatment preferences for depression in epilepsy. *Epilepsy & Behavior* 2011; 22: 740–744.
6 Cathebras P, Mosnier C, Levy M, Bouchou K, Rousset H. Screening for depression in patients with medical hospitalization. Comparison of two self-evaluation scales and clinical assessment with a structured questionnaire. *L'Encéphale* 1994; 20: 311–317.
7 Gemmill AW, Leigh B, Ericksen J, Milgrom J. A survey of the clinical acceptability of screening for postnatal depression in depressed and non-depressed women. *BMC Public Health* 2006; 6: 211–218.
8 Rees G, Holloway EE, Craig G et al. Screening for depression: Integrating training into the professional development programme for low vision rehabilitation staff. *Clinical & Experimental Ophthalmology* 2012; 40: 840–848.
9 Kroenke K, Spitzer R, Williams JBW. The Patient Health Questionnaire-2: Validity of a two-item depression screener. *Medical Care* 2003; 41: 1284–1292.
10 Randall J, Voth R, Burnett E, Bazhenova L, Bardwell W. Clinic-based depression screening in lung cancer patients using the PHQ-2 and PHQ-9 depression questionnaires: A pilot study. *Supportive Care in Cancer* 2013; 21: 1503–1507.
11 Sowden G, Mastromauro CA, Januzzi JL, Fricchione GL, Huffman JC. Detection of depression in cardiac inpatients: Feasibility and results of systematic screening. *American Heart Journal* 2010; 159: 780–787.
12 Li C, Friedman B, Conwell Y, Fiscella K. Validity of the Patient Health Questionnaire 2 (PHQ-2) in identifying major depression in older people. *Journal of the American Geriatrics Society* 2007; 55: 596–602.
13 Mitchell AJ. Clinical utility of screening for clinical depression and bipolar disorder. *Current Opinion in Psychiatry* 2012; 25: 24–31.
14 US Preventive Services Task Force. Screening for depression in adults: US Preventive Services Task Force recommendation statement. *Annals of Internal Medicine* 2009; 151: 784–792.
15 Zivin K, Kales HC. Adherence to depression treatment in older adults: A narrative review. *Drugs & Ageing* 2008; 25: 559–571.
16 Swift JK, Callahan JL. The impact of client treatment preferences on outcome: A meta-analysis. *Journal of Clinical Psychology* 2009; 65: 368–381.
17 Holloway EE, Sturrock BA, Lamoureux EL, Keeffe JE, Rees G. Help seeking among vision impaired adults referred to their GP for depressive symptoms: patient characteristics and outcomes associated with referral uptake. *Australian Journal of Primary Health* 2013. doi: 10.1071/PY13085 [Epub ahead of print].
18 Brooke P, Bullock R. Validation of a 6 item cognitive impairment test with a view to primary care usage. *International Journal of Geriatric Psychiatry* 1999; 14: 936–940.
19 Kroenke K, Spitzer RL, Williams JBW. The PHQ-9. *Journal of General Internal Medicine* 2001; 16: 606–613.
20 Kroenke K, Spitzer RL, Williams JBW, Löwe B. The Patient Health Questionnaire somatic, anxiety, and depressive symptom scales: A systematic review. *General Hospital Psychiatry* 2010; 32: 345–359.
21 Zimmerman M, Farber NJ, Hartung J, Lush DT, Kuzma MA. Screening for psychiatric disorders in medical patients: A feasibility and patient acceptance study. *Medical Care* 1994; 32: 603–608.
22 Gum AM, Arean PA, Hunkeler E et al. Depression treatment preferences in older primary care patients. *The Gerontologist* 2006; 46: 14–22.
23 Hayman KJ, Kerse NM, La Grow SJ, Wouldes T, Robertson MC, Campbell AJ. Depression in older people: Visual impairment and subjective ratings of health. *Optometry and Vision Science* 2007; 84: 1024–1030.
24 Bookwala J, Lawson B. Poor vision, functioning, and depressive symptoms: A test of the activity restriction model. *The Gerontologist* 2011; 51: 798–808.
25 Mohr DC, Ho J, Duffecy J et al. Perceived barriers to psychological treatments and their relationship to depression. *Journal of Clinical Psychology* 2010; 66: 394–409.
26 Nguyen HV, Inderjeeth CA, Tang E, Barnabas L, Merriam M. Screening for depression in hospitalised and community-dwelling elderly: The use of the 4-item, 5-item and 15-item geriatric depression scales. *Australasian Journal on Ageing* 2006; 25: 204–208.
27 Arroll B, Goodyear-Smith F, Crengle S et al. Validation of PHQ-2 and PHQ-9 to screen for major depression in the primary care population. *The Annals of Family Medicine* 2010; 8: 348–353.
28 Lichtman JH, Bigger JT Jr, Blumenthal JA et al. Depression and coronary heart disease: Recommendations for screening, referral, and treatment. *Circulation* 2008; 118: 1768–1775.
29 Gjerdingen D, Crow S, McGovern P, Miner M, Center B. Postpartum depression screening at well-child visits: validity of a 2-question screen and the PHQ-9. *The Annals of Family Medicine* 2009; 7: 63–70.
30 Rees G, Mellor D, Holloway E et al. Integrated depression management: A proposed trial of a new model of care in a low vision rehabilitation setting. *Ophthalmic Epidemiology* 2013; 20: 321–329.

Appendix I

The Patient Health Questionnaire-2 (PHQ-2) depression screening test

Over the past 2 weeks, on a scale of 0–3, how often have you been bothered by any of the following problems:

1. Feeling down, depressed or hopeless
2. Little interest or pleasure in doing things

© 2014 ACOTA